Xi Li

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Profile

Ph.D. researcher in Computer Science at UC Irvine. Trained in **core computer vision at Tsinghua University** before expanding into biomedical research during my Ph.D., bringing complementary perspectives in visual representation learning and biological problem formulation. My work connects tissue morphology, spatial gene expression, and cellular context through multimodal learning and reliable evaluation. First author of **sMMC-22M** (**NeurIPS 2026**, accepted poster) and **STMoE** (**CIKM 2026**, accepted); co-author of **TargetSage** (**NeurIPS 2026**, accepted poster) and co-first author of **MUSE** (**CIKM 2025**).

Education

University of California, Irvine *Sep 2023 – Present*
Ph.D., Computer Science | **Expected graduation: December 2027**

Tsinghua University *Sep 2019 – Jun 2022*
M.S., Computer Science and Technology

University of Science and Technology of China *Sep 2015 – Jun 2019*
B.S., Information Security

Research Experience

University of California, Irvine *Sep 2023 – Present*
Ph.D. Student Researcher | Irvine, CA

- **sMMC-22M | Multimodal data and benchmarks** *(first author; **NeurIPS 2026**, accepted poster).* Built a context-aware resource linking cell morphology, molecular profiles, and tissue context; the expanded collection contains **28.3M cell-aligned targets across 25 tissue categories**. Tracked study, assay, and coordinate provenance, explicitly distinguishing native Xenium cells from derived Visium HD targets. Designed **provenance-aware data curation and evaluation** to study how image-to-expression models transfer across samples and platforms. The paper retains the frozen-release name sMMC-22M.

- **TargetSage | Interpretable therapeutic target discovery** *(co-author; **NeurIPS 2026**, accepted poster).* Responsible for **disease-state verification and multi-target investigation** in a framework that combines retrieved biomedical evidence, gene-specific LLM reasoning, and positive-unlabeled learning. This formulation avoids treating potentially druggable but unlabeled genes as confirmed negatives. The overall model improved **macro-average Adjusted F1 by 37%** over the strongest of 10 baselines across **15 tasks and 19,032 genes**. Used Perturb-seq data to fine-tune scGPT in TargetSage and InverseTURB-related research, linking disease-state context to cellular responses.

- **STMoE | Histology-to-expression learning** *(first author; **CIKM 2026**, accepted).* Led a multi-scale mixture-of-experts model for **single-cell gene-expression prediction across 25 organs**. Combined cell and tissue-context crops, frozen UNI2-h representations, soft contrastive alignment, and organ-routed retrieval with training-only molecular reference banks; retained **70% of predictive performance** under cross-sample generalization.

- **MUSE | Multi-slice spatial transcriptomics** *(co-first author; **CIKM 2025**).* Co-developed joint analysis of spatial transcriptomics experiments to capture information shared across tissue slices and complement single-sample modeling.

- **Spine | Context-aware spatial response modeling** *(first author; **KDD 2027**, under review).* Built **Pre-Post Perturbation Pairing (PPP)**, an audited matched-control pseudo-target benchmark, and a **two-stage graph model** that freezes tissue-context predictions before routing direct-effect profiles from perturbed cells to predict residual neighbor responses. On **1,342/837/928 spatially disjoint train/validation/test graphs**, raised pooled Ring Pearson from **0.3626 to 0.4111** over context-only prediction (partial $R^2 = 0.1521$). A validation-fixed selector preserved target-qualified rate (0.6944) while reducing mean transcriptomic spillover by **12.0%**. Gene-disjoint matching and held-out-guide audits separated local perturbation signal from context and identified cross-assay direct-effect transfer as a bottleneck, supporting target-retaining, lower-burden experimental prioritization.

- **Biomedical foundation-model bias and reliability** *(first author; **KDD 2027**, under review).* Developed *Context Is Not a Side Note*, a support-calibrated framework auditing **frozen single-cell and pathology embeddings** through average/worst-context performance, context gaps, and donor/patient support. Built a **label-context support graph** to gate reweighting (LC-Reweight) and conditional adaptation (SCA-Align), distinguishing observed-context interpolation from held-out assay/dataset generalization. Detected context-specific failures in **43/50 eligible evaluation rows** (median gap 0.404) using clustered-bootstrap/FDR screening. In a held-out Geneformer case, SCA-Align reduced neutrophil-abundance contrast distortion by **31%** and recovered its mean direction. Cross-tissue benchmarks established task-dependent gains and explicit limits where label-context support is insufficient.

Mentoring and Open-Source Leadership

Google Summer of Code (GSoC) *2025 & 2026*

Project Mentor. Mentored two students through problem formulation, implementation, experiments, and paper development; both projects have code repositories. Guided one student-first-author paper at **BIBM**. **2026: Omni-ST**, instruction-driven multimodal spatial modeling. **2025: Disentangled Generation and Editing of Pathology Images**.

Earlier Research and Industry Experience

Tsinghua University & Eastern Institute of Technology, Ningbo

Research Assistant (concurrent appointments) *Oct 2021 – Sep 2022*

Worked at Tsinghua's Multimedia Lab and Eastern Institute of Technology under the same advisor, who held appointments at both institutions. Collaborated with China's National Centre for the Performing Arts on multiple projects, including AI-powered multi-camera directing, and with Alibaba on AI-powered livestream commerce. Pursued an AI imaging startup venture with my advisor, leading product definition and delivery across research, engineering, and design.

ByteDance AI Lab *May 2021 – Dec 2021*

AI Product Manager – Full-Time Intern. Incubated **Alchemy** for Fanqie Novel book covers; supported **Chimera** assistant solutions through **XPENG/QCraft** client engagement.

ByteDance – Feishu/Lark *Jun 2020 – Nov 2020*

Data Product – Full-Time Intern. Applied behavior and relevance analysis to enterprise search-product design.

Nolibox / Computational Aesthetics *Mar 2020 – Aug 2020*

Partner and Co-founder. Built automatic logo-generation and font-stroke generation tools for AI-assisted design.

Selected Publications and Manuscripts

Full list and links: xenali96.github.io/publications.html

Published / accepted

1. sMMC-22M: A Context-Aware Dataset and Benchmark for Single-Cell Spatial Transcriptomics.
Xi Li et al. | **NeurIPS 2026**, accepted poster. *First author.*

2. TargetSage: Identifying Therapeutic Target Genes with Interpretable and Robust LLM Reasoning.
Z. Duan, T. Wu, **Xi Li**, et al. | **NeurIPS 2026**, accepted poster.

3. STMoE: Multi-Scale Mixture-of-Experts for Single-Cell Gene Expression Prediction from Histology.
Xi Li et al. | **CIKM 2026**, accepted. *First author.*

4. MUSE: A Multi-Slice Joint Analysis Method for Spatial Transcriptomics Experiments.
Z. Duan*, **Xi Li***, et al. | **CIKM 2025**. *Co-first author.*

5. iHAST: Integrating Hybrid Attention for Super-Resolution in Spatial Transcriptomics.
Xi Li et al. | **BMVC 2024**. *First author.*

6. DISCO: A Diffusion Model for Spatial Transcriptomics Data Completion.
Z. Duan, **Xi Li**, et al. | **ICIP 2025**.

Manuscripts

7. Context Is Not a Side Note: A Support-Calibrated Reliability Framework for Biomedical Foundation Models.
Xi Li et al. | **KDD 2027**, under review. *First author.*

8. Spine: Disentangling Tissue Context and Perturbation-Conditioned Response Fields in Spatial Genomics.
Xi Li et al. | **KDD 2027**, under review. *First author.*

9. Spatially Indexed Single-Cell Morphology and Transcriptome Profiling by DBiC-seq.
L. Lin, S. Bao, **Xi Li**, et al. | **Nature Communications**, under review.

10. CCR2 Deficiency Attenuates Flow-Induced Carotid Artery Remodeling.
Vascular Pharmacology, under review. *Co-author; corresponding author: Yang Liu.*

11. Cross-Species Multi-Omics Identifies a Conserved Fibrogenic Valvular Interstitial Cell Population across Mitral Valve Diseases.
F. Gao et al. (including **Xi Li**) | **Circulation**, submitted.

Technical Skills

Biomedical ML: Computational pathology; H&E/whole-slide imaging; spatial transcriptomics; single-cell genomics; image-omics alignment; Perturb-seq/scGPT adaptation.

Modeling: Multimodal transformers; mixture-of-experts; contrastive learning; retrieval-augmented prediction; LLM reasoning; population/context reliability.

Engineering: Python; PyTorch; multi-GPU experimentation; large-scale data curation and QC; patient/study/assay-aware benchmarking.